

Mark the following Statements as True or False (Use T for True, F for False):

- i. $2 \in \{x \in \mathbf{R} \mid x \text{ is an integer greater than } 1\}$ _____
- ii. $2 \in \{x \in \mathbf{R} \mid x \text{ is the square of an integer}\}$ _____
- iii. $2 \in \{2, \{2\}\}$ _____
- iv. $2 \in \{\{2\}, \{\{2\}\}\}$ _____
- v. $2 \in \{\{2\}, \{2, \{2\}\}\}$ _____
- vi. $2 \in \{\{\{2\}\}\}$ _____
- vii. $\{2\} \in \{x \in \mathbf{R} \mid x \text{ is an integer greater than } 1\}$ _____
- viii. $\{2\} \in \{x \in \mathbf{R} \mid x \text{ is the square of an integer}\}$ _____
- ix. $\{2\} \in \{2, \{2\}\}$ _____
- x. $\{2\} \in \{\{2\}, \{\{2\}\}\}$ _____
- xi. $\{2\} \in \{\{2\}, \{2, \{2\}\}\}$ _____
- xii. $\{2\} \in \{\{\{2\}\}\}$ _____
- xiii. $\{1, 3, 3, 3, 5, 5, 5, 5, 5\} = \{5, 3, 1\}$ _____
- xiv. $\{\{1\}\} = \{1, \{1\}\}$ _____
- xv. $\emptyset = \{\emptyset\}$ _____
- xvi. $x \in \{x\}$ _____
- xvii. $\{x\} \subseteq \{x\}$ _____
- xviii. $\{x\} \in \{x\}$ _____
- xix. $\{x\} \in \{\{x\}\}$ _____
- xx. $\emptyset \subseteq \{x\}$ _____
- xxi. $\emptyset \in \{x\}$ _____

Use a Venn diagram to illustrate the relationship $A \subseteq B$ and $B \subseteq C$.

Use a Venn diagram to illustrate the relationships $A \subset B$ and $B \subset C$.

What is the cardinality of each of these sets?

- i. $\emptyset$ _____
- ii. $\{\emptyset\}$ _____
- iii. $\{\emptyset, \{\emptyset\}\}$ _____
- iv. $\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$ _____
- v. $\mathcal{P}(\{a, b, \{a, b\}\})$ _____
- vi. $\mathcal{P}(\{\emptyset, a, \{a\}, \{\{a\}\}\})$ _____
- vii. $\mathcal{P}(\mathcal{P}(\emptyset))$ _____
- viii. $\{a, b, c, d, e\} \times \{y, z\}$ _____
- ix. $\{y, z\} \times \{a, c, d\}$ _____
- x. $\{a, b, c\} \times \{x, y\} \times \{0, 1\}$ _____

Find the truth set of each of these predicates where the domain is the set of integers.

- i. $P(x): x^2 < 3$ { _____ }
- ii. $Q(x): x^2 > x$ { _____ }
- iii. $R(x): 2x + 1 = 0$ { _____ }
- iv. $P(x): x^3 \geq 1$ { _____ }
- v. $Q(x): x^2 = 2$ { _____ }

vi. $R(x): x < x^2$ { }

Show that $(A - B) - C = (A - C) - (B - C)$.

$$(A - C) - (B - C) \equiv (A \cap \overline{C})$$

Draw the Venn diagrams for each of these combinations of the sets A , B , and C .

$$A \cap (B \cup C)$$

$$(A - B) \cup (A - C) \cup (B - C)$$

$$A \cap B) \cup (A \cap C)$$

What can you say about the sets A and B if we know that

a) $A \cup B = A$?

b) $A \cap B = A$?

c) $A - B = A$?

d) $A \cap B = B \cap A$?

e) $A - B = B - A$?

Draw a Venn diagram for the symmetric difference of the sets A and B .

What can you say about the sets A and B if $A \oplus B = A$?

Let $A_i = \{\dots, -2, -1, 0, 1, \dots, i\}$. Then

$$\bigcup_{i=1}^n A_i = \{ \quad \quad \quad \}$$

$$\bigcap_{i=1}^n A_i = \{ \quad \quad \quad \}$$

Suppose that the universal set is $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Find the set specified by each of these bit strings.

a) 11 1100 1111 { }

b) 01 0111 1000 { }

c) 10 0000 0001 { }

Given two sets A and B (as bit strings), what is the bit string corresponding to

$$A - B$$

$$B - A$$

$$A \cap B$$

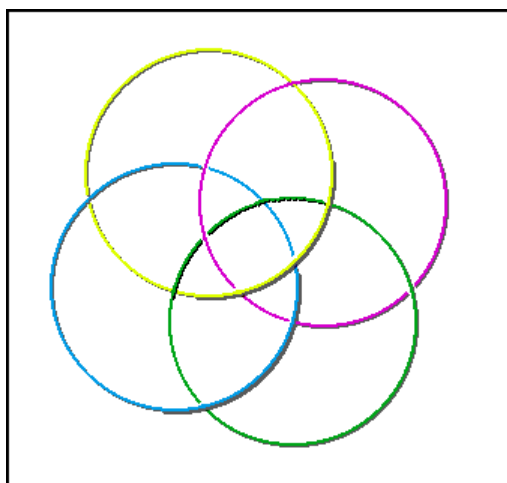
$$A \cup B$$

Given two sets A and B (as bit strings), how can we determine that

$$A \subseteq B?$$

$$A \subset B?$$

Why the following figure is not a Venn Diagram for 4 sets?



Given the sets $B=\{1,2,3,4,5\}$, $C=\{3,5,7\}$, $D=\{2,4,6,8\}$, and $E=\{a,b\}$; **what would be the set A if:**

$$|A| = 4, |A \cap B| = 2, |A \cap C| = 2, A - D = A, \text{ and } A \cap E = E$$

i) Given three sets A, B, and C; if $A \cap C = B \cap C$, then must it hold that $A=B$? Justify your answer with adequate reasoning and example.

Hint: The answer is NO; just give a counterexample (i.e. Three sets A, B and C, such that $A \cap C = B \cap C$ but $A \neq B$).

ii) What can you say (about $A=B$), if $A \cup C = B \cup C$? Give reasoning (and example).

Hint: The answer is NO; just give a counterexample (i.e. Three sets A , B and C , such that $A \cup C = B \cup C$ but $A \neq B$).

iii) What can you say (about $A=B$), if $A \cup C = B \cup C$ and $A \cap C = B \cap C$? Give reasoning (and example).

Hint: The answer is YES. Justify your answer by establishing that $A \subseteq B$ and $B \subseteq A$.

Show that $A - B = A \cap \bar{B}$ using set builder notation.

Let A be the set of students having home in Lahore, let B be the set of students of FAST (Lahore), and let C be the set of students living in different hostels of Lahore; **then what do the following sets represent?**

i) $((A' \cup C) - B)'$

- Here A' means, *complement* of the set A ; similarly, $((A' \cup C) - B)'$ means, *complement* of the set $((A' \cup C) - B)$.

ii) $((A \cap B) - C)'$

Note: The universal set consists of all students of BSSE studying in different institutions of Lahore.

Why is f not a function from $\mathbb{R}$ to $\mathbb{R}$ if

$$f(x) = 1/x?$$

$$f(x) = \sqrt{x}?$$

$$f(x) = \pm\sqrt{(x^2 + 1)}?$$

Determine whether f is a function from $\mathbb{Z}$ to $\mathbb{R}$ if

$$f(n) = \pm n.$$

$$f(n) = \sqrt{n^2 + 1}.$$

$$f(n) = 1/(n^2 - 4).$$

Determine whether f is a function from the set of all bit strings to the set of integers if

$f(S)$ is the position of a 0 bit in S .

$f(S)$ is the number of 1 bits in S .

$f(S)$ is the smallest integer i such that the i th bit of S is 1 and $f(S) = 0$ when S is the empty string, the string with no bits.

Determine whether $f: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ is onto if

$$f(m, n) = 2m - n.$$

$$f(m, n) = m^2 - n^2.$$

$$f(m, n) = m^2 + n^2.$$

$$f(m, n) = m + n + 1.$$

$$f(m, n) = |m| - |n|.$$

$$f(m, n) = m^2 - 4.$$

Determine whether each of these functions from $\mathbf{Z}$ to $\mathbf{Z}$ is one-to-one.

$$f(n) = n - 1$$

$$f(n) = n^2 + 1$$

$$f(n) = n^3$$

$$f(n) = \lceil n/2 \rceil$$

Give an example of an increasing function from $\mathbf{R}$ to itself

Give an example of a strictly increasing function from $\mathbf{R}$ to itself

Give an example of a decreasing function from $\mathbf{R}$ to itself

Give an example of a strictly decreasing function from $\mathbf{R}$ to itself

Let $S = \{-1, 0, 2, 4, 7\}$. Find $f(S)$ if

$$f(x) = 1. \quad \{ \quad \}$$

$$f(x) = 2x + 1. \quad \{ \quad \}$$

$$f(x) = \lceil x/5 \rceil. \quad \{ \quad \}$$

$$f(x) = \lfloor (x^2 + 1)/3 \rfloor. \quad \{ \quad \}$$

Let $f(x) = \lfloor x^2/3 \rfloor$. Find $f(S)$ if

$$S = \{-2, -1, 0, 1, 2, 3\}. \quad f(S) = \{ \quad \}$$

$$S = \{0, 1, 2, 3, 4, 5\}. \quad f(S) = \{ \quad \}$$

$$S = \{1, 5, 7, 11\}. \quad f(S) = \{ \quad \}$$

$$S = \{2, 6, 10, 14\}.$$

$$f(S) = \{ \quad \quad \quad \}$$

Find $f \circ g$ and $g \circ f$, where $f(x) = x^2 + 1$ and $g(x) = x + 2$, are functions from $\mathbb{R}$ to $\mathbb{R}$.

Show that if x is a real number and m is an integer, then $\lceil x + m \rceil = \lceil x \rceil + m$

Show that if x is a real number, then $x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1$.

Show that if x is a real number and n is an integer, then

$x < n$ if and only if $\lfloor x \rfloor < n$.

$n < x$ if and only if $n < \lceil x \rceil$.

Prove that if x is a real number, then $\lfloor 2x \rfloor = \lfloor x \rfloor + \left\lfloor x + \frac{1}{2} \right\rfloor$.

Prove or disprove that $\lceil x + y \rceil = \lceil x \rceil + \lceil y \rceil$ for all real numbers x and y .